

Education

2025 - 2027 **BSc. in Applied Mathematics and Statistics**, *University of Massachusetts Lowell*
Minor in Computer Science

Experience

- June 2026–present **Undergraduate Research Fellow**, *Department of Mathematics and Statistics*, University of Massachusetts Lowell, Advisor: Shiwen Zhang, PhD
- Conduct independent study on random walks, discrete diffusion, and their connections to the heat equation.
 - Study lattice-based diffusion models and their applications to heterogeneous biological systems.
 - Support mathematical and computational investigation of diffusion behavior using analytical and numerical methods.
- June 2025–present **Research Assistant in Biomedical Data Science**, *Department of Biological Sciences*, University of Massachusetts Lowell, Advisor: Rachel Melamed, PhD
- Apply statistical and data analysis methods to investigate how genetic variants contribute to Alzheimer's disease-related traits.
 - Process and analyze genomic and clinical datasets from GTEx and ADNI using Python, Bash, and bioinformatics tools.
 - Apply PrediXcan-based gene expression prediction models to study associations between genotype, predicted gene expression, and clinical phenotypes.
 - Prepare cleaned and structured datasets for downstream machine learning and statistical analysis.
- Jan 2026–present **Mathematics Tutor**, *Department of Mathematics and Statistics*, University of Massachusetts Lowell
- Tutor undergraduate students in a variety of mathematics courses through walk-in support services.
 - Clarify course material by breaking down complex problems and reinforcing key mathematical concepts.
 - Help students develop stronger analytical skills, improve homework performance, and prepare for exams.

Awards and Honors

- 2026 UMass Lowell Kennedy Summer Scholarship
2025 UMass Lowell Moloney Scholarship
2025 UMass Lowell International Transfer Academic Achievement Award

Technical Skills

- Languages** Python, R, Bash, SQL, Java, C++
Libraries pandas, NumPy, scikit-learn, matplotlib
Bioinformatics bcftools, vcftools, ANNOVAR, PrediXcan, VCF files
Tools Git, GitHub, Linux, LaTeX

Relevant Coursework

Mathematics Applied Linear Algebra, Optimization, Real Analysis, Partial Differential Equations, Applied Statistics, Probability and Mathematical Statistics, Math methods for Quantum error correction (audit)

Computer Science Java Programming, C++ Programming, SQL Programming, Data Structures & Algorithms, Organization of Programming Languages, Data Science for Biologists

Presentation

Mar 2026 **Student Research and Community Engagement Symposium**, UMass Lowell, Oral presentation
Project: Understand the effect of genetics variants on Alzheimer's Disease